

well as size, color, and odor. Much of this diversity results from adaptation to specific pollinators (see Figures 38.4 and 38.5).

Fruits

As seeds develop from ovules after fertilization, the ovary wall thickens and the ovary matures into a **fruit**. A pea pod is an example of a fruit, with seeds (mature ovules, the peas) encased in the ripened ovary (the pod).

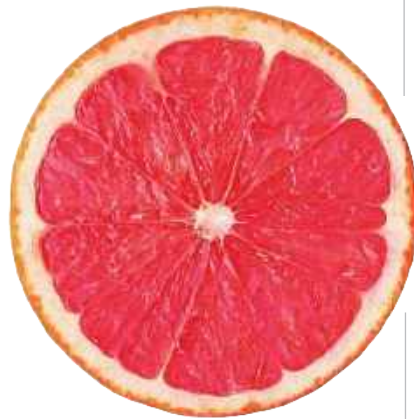
Fruits protect seeds and aid in their dispersal. Mature fruits can be either fleshy or dry (**Figure 30.10**). Tomatoes, plums, and grapes are examples of fleshy fruits, in which the wall (pericarp) of the ovary becomes soft during ripening. Dry fruits include beans, nuts, and grains. Some dry fruits split open at maturity to release seeds, whereas others remain closed. The dry, wind-dispersed fruits of grasses, harvested while on the plant, are major staple foods for humans. The cereal grains of maize, rice, wheat, and other grasses, though easily mistaken for seeds, are each actually a fruit with a dry outer covering (the former wall of the ovary) that adheres to the seed coat of the seed within.

▼ **Figure 30.10** Some variations in fruit structure.

- ▼ Tomato, a fleshy fruit with soft outer and inner layers of pericarp (fruit wall)



- ▼ Ruby grapefruit, a fleshy fruit with a firm outer layer and soft inner layer of pericarp



- ▼ Nectarine, a fleshy fruit with a soft outer layer and hard inner layer (pit) of pericarp



- ▼ Hazelnut, a dry fruit that remains closed at maturity



- ▼ Milkweed, a dry fruit that splits open at maturity



As shown in **Figure 30.11**, various adaptations of fruits and seeds help to disperse seeds (see also Figure 38.12). The seeds of some flowering plants, such as dandelions and maples, are contained within fruits that function like parachutes or propellers, adaptations that enhance dispersal by wind. Some fruits, such as coconuts, are adapted to dispersal by water. And the seeds of many angiosperms are carried by animals. Some angiosperms have fruits modified as burrs that cling to animal fur (or the clothes of humans). Others produce edible fruits, which are usually nutritious, sweet tasting, and vividly colored, advertising their ripeness. When an animal eats the fruit, it digests the fruit's fleshy part, but the tough seeds usually pass unharmed through the animal's digestive tract. When the animal defecates, it may deposit the seeds, along with a supply of natural fertilizer, many kilometers from where the fruit was eaten.

▼ **Figure 30.11** Fruit adaptations that enhance seed dispersal.



- Some plants have mechanisms that disperse seeds by explosive action.

- Wings enable maple fruits to be carried by the wind.



- Seeds within berries and other edible fruits are often dispersed in animal feces.

- The barbs of cockleburs facilitate seed dispersal by allowing the fruits to "hitchhike" on animals.



➔ **Mastering Biology Animation: Fruit Structure and Seed Dispersal**